

Duke Energy Corporation

Largest US Regulated Electric Utility — AI Datacentre Baseload Demand Play

PhD Due Diligence Report | BLRI-DD-DUK-AUG2026

13 August 2026 | Blue Lotus Intelligence | CivilisationOS

Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

Duke Energy: The Backbone of Southeast US Power Infrastructure

Duke Energy serves 8.4 million electric customers across the Carolinas, Florida, Indiana, Ohio, and Kentucky — the largest regulated electric utility in the United States by customer count. In Seldon ξ -variable terms, Duke is a civilisational energy distribution infrastructure company: it controls the transmission and distribution of electricity to one of the fastest-growing regions in the US, including the Research Triangle (NC) and Charlotte AI/fintech corridor. The AI datacentre buildout in Duke's Carolinas territory is accelerating: Microsoft, Apple, Google, and Meta have all announced hyperscale campus projects in Duke's service territory, driving unprecedented load growth.

Duke's regulated utility model provides a Seldon Ω -reducing function: rate-regulated monopoly returns are predictable, providing energy supply security for millions of households. The risk is rate case timing — Duke must file for rate increases with state commissions to recover capital investments in grid modernisation and new generation, creating regulatory lag between investment and return.

Source: moomoo OpenD [2026-08-13] · Duke Energy investor relations · NC/FL utility commission filings

PART II — QUALITATIVE ANALYSIS

Carolinas AI Corridor: Duke's Fastest-Growing Load

The Research Triangle Park (Raleigh-Durham) and Charlotte metro are among the fastest-growing data centre markets in the US. Duke's Carolinas Carolinas subsidiaries (Duke Energy Carolinas + Duke Energy Progress) face >4 GW of new datacentre load requests in their interconnection queues — a 10× increase since 2022. Duke is accelerating its grid modernisation capex programme (>\$65B over 10 years) to serve this demand, funded through rate cases.

Nuclear Fleet: Highest Capacity Factor in Duke's Generation Mix

Duke operates the largest nuclear fleet of any US utility — 11 nuclear units across 6 plants, providing ~10 GW of firm, carbon-free baseload. This nuclear fleet is the key competitive advantage vs. peers: nuclear operates at >90% capacity factor, providing the most reliable baseload generation available. Duke's nuclear fleet is fully amortised, making it the lowest marginal-cost generation in its mix.

Source: Duke Energy 10-K EDGAR · EEI utility comparison · NRC reactor status

PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]

Price: \$124.590 | Prev Close: \$123.490 | Mkt Cap: \$97.143B

EPS TTM: \$6.310 | Net Income: \$4.920B | Div Yield: 3.42%

52w: \$111.966–\$133.338 | PE TTM: 18.71x | Shares: 779.7M

Metric	Value	Source
Price (2026-08-13)	\$124.590	moomoo OpenD
Market Cap	\$97.143B	moomoo OpenD
EPS TTM	\$6.310	moomoo OpenD
Net Income TTM	\$4.920B	moomoo OpenD
52-Week Range	\$111.966–\$133.338	moomoo OpenD
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PART IV — COMPETITIVE LANDSCAPE

Competitor	Type	Mkt Cap	vs DUK
NextEra Energy (NEE)	Regulated utility + renewables	\$179.6B	NEE larger; DUK has nuclear advantage
Southern Company (SO)	Regulated utility (Southeast)	~\$90B	DUK larger; Vogtle nuclear delayed vs DUK fleet
Dominion Energy (D)	Mid-Atlantic regulated utility	~\$45B	DUK larger; DUK Carolinas AI corridor more dynamic
Entergy (ETR)	Gulf Coast regulated utility	~\$27B	DUK 3× larger; DUK nuclear fleet superior

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The Microsoft Datacentre Site Selection VP (COMPOSITE)

A Microsoft real estate executive evaluates Duke's North Carolina service territory for a 500 MW hyperscale campus. 'Duke's ability to offer 20-year power purchase agreements backed by their nuclear fleet is the deciding factor over other Southeast utilities. The Research Triangle grid is mature enough for Phase 1 build-out.' Represents the hyperscale datacentre demand wave that is transforming Duke's load growth from 1% to 3%+ annually.

COMPOSITE / ILLUSTRATIVE — drawn from datacentre site selection patterns

FINDING F-5-B: Portrait 2: The North Carolina Utility Commissioner (COMPOSITE)

A state utility commissioner oversees Duke's rate case in Raleigh. 'We must balance ratepayer protection against Duke's legitimate capital recovery needs. The datacentre load boom is real, but residential customers must not subsidise corporate electricity users.' Represents the regulatory risk: rate case outcomes determine Duke's allowed ROE and capex recovery timeline — the core uncertainty in the utility investment thesis.

COMPOSITE / ILLUSTRATIVE — drawn from state utility regulation patterns

FINDING F-5-C: Portrait 3: The Duke Nuclear Plant Operator (COMPOSITE)

A senior reactor operator at McGuire Nuclear Station (Charlotte, NC) notes: 'Our fleet is running at 93% capacity factor. We're the most reliable generation asset in the Duke system — zero carbon, zero fuel price risk. The AI datacentres want our power directly.' Represents the nuclear baseload advantage: fully amortised, lowest marginal cost, 24/7 firm power — exactly what datacentre customers require.

COMPOSITE / ILLUSTRATIVE — drawn from US nuclear plant operations patterns

PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: Rate case denial squeezes ROE below cost of capital

EVIDENCE FOR:

Duke's Carolinas rate cases have been contested by industrial customers and consumer advocates; commissions may disallow portions of grid modernisation capex

EVIDENCE AGAINST:

Duke's regulated monopoly status means some ROE is always guaranteed; datacentre load growth strengthens Duke's case for rate increases

QUANTITATIVE TEST

$P(\text{material rate case denial reducing ROE below 9\% by 2027}) = 20\%$

VERDICT: BEAR — regulatory timing mismatch is Duke's primary risk

Implication: Monitor NC/FL/IN rate case dockets; track allowed ROE trend vs. peer utilities

HH2: HH2: AI datacentre load exceeds grid capacity — Duke wins pricing power

EVIDENCE FOR:

Datacentre queue >4 GW; transmission congestion in Research Triangle; Duke can negotiate premium bilateral contracts above regulated rates

EVIDENCE AGAINST:

Regulated utility model limits pricing above commission-approved rates; load growth benefits take years to flow through rate cases

QUANTITATIVE TEST

P(DUK earning above-regulated-returns from datacentre direct contracts by 2027)=15%

VERDICT: BULL — bilateral contract premium would re-rate DUK above utility peers

Implication: Monitor Duke's IRP filings and any bilateral contract announcements with hyperscalers

HH3: HH3: Nuclear relicensing delays create generation gap

EVIDENCE FOR:

Oconee Nuclear (3 × 900 MW) licence renewal due 2033; regulatory delays possible; gap in baseload creates reliability risk

EVIDENCE AGAINST:

NRC has approved licence renewals for 20+ US reactors; Duke has strong nuclear operations track record; 20-year extensions are standard

QUANTITATIVE TEST

P(material nuclear unit closure before replacement capacity online)=10%

VERDICT: BEAR-NEUTRAL — nuclear risk is long-duration and manageable

Implication: Monitor NRC licence renewal dockets for Duke's six nuclear plants

HH4: HH4: Southeast load growth drives multi-decade capex supercycle

EVIDENCE FOR:

Population migration to Carolinas/Florida accelerating; EV charging load + datacentre load + residential growth = 3%+ annual load growth through 2040; Duke's capex programme is a multi-decade earnings growth engine

EVIDENCE AGAINST:

Capex requires equity issuance that dilutes existing shareholders; construction risk on new generation projects

QUANTITATIVE TEST

P(DUK achieving 5%+ EPS CAGR through 2030)=55%

VERDICT: BULL — load growth tailwind is the strongest utility bull case in decades

Implication: Monitor quarterly load growth data; track IRP capex authorisation

HH5: HH5: Interest rate sensitivity compresses utility valuation multiples

EVIDENCE FOR:

Regulated utilities are bond proxies; rising interest rates increase discount rate and compress P/E; Duke's 3.42% yield becomes less attractive vs. rising risk-free rate

EVIDENCE AGAINST:

Duke's 5%+ dividend growth guidance maintains real yield; rate-regulated earnings are non-cyclical and inflation-passthrough mechanisms exist

QUANTITATIVE TEST

P(DUK P/E compressing below 15× if 10yr Treasury >5%)=30%

VERDICT: BEAR — rate sensitivity is the primary macro risk for all utility stocks

Implication: Monitor 10-year Treasury yield; Duke's dividend growth vs. risk-free rate spread

PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: Carolinas datacentre load drives 3%+ EPS growth (P = 55%)

Duke files successful rate cases; datacentre load growth 4GW by 2028; EPS grows to \$7.50+; stock re-rates to \$140

Driver: Quarterly load growth data; rate case approval timelines

Falsifier: Rate case denial; commission limits datacentre cost recovery

SCENARIO 2: Nuclear fleet extension provides 20-year baseload advantage (P = 65%)

Oconee/McGuire/Catawba licence renewals approved; nuclear fleet runs to 2053; lowest-cost generation baseline secured

Driver: NRC licence renewal filings; Duke IRP nuclear section

Falsifier: NRC denies renewal; forced premature shutdown of units

SCENARIO 3: Base case: 4-5% dividend growth, stock range \$115-135 (P = 60%)

Regulated utility compounding; 3.42% yield + 4-5% dividend growth = 7-9% total return; stock tracks earnings growth

Driver: Quarterly EPS; dividend increase announcements

Falsifier: Adverse rate case + rising rates compresses multiple

SCENARIO 4: Hyperscaler direct PPA at premium rates (P = 20%)

Microsoft/Google signs 20-year direct power purchase agreement above regulated rates; Duke's earnings above regulated returns; analyst upgrade cycle

Driver: PPA announcement; bilateral contract filing with FERC

Falsifier: Commission prohibits bilateral contracts above regulated tariff

SCENARIO 5: Florida growth market outperforms Carolinas (P = 25%)

Duke Energy Florida (Tampa/St. Pete territory) serves residential + commercial growth; IRP expansion; Florida rate cases historically constructive

Driver: Florida PSC rate case outcomes; FPL comparisons

Falsifier: Florida rate case cut allowed ROE; residential opposition

SCENARIO 6: Interest rate compression drives stock to \$110 (P = 25%)

10-year Treasury spikes to 5.5%; utility sector selloff; DUK de-rates to 16x earnings; tests \$108 support

Driver: Treasury yield; utility ETF fund flows

Falsifier: Fed rate cut cycle accelerates; yields fall, utilities re-rate up

SCENARIO 7: Grid modernisation capex drives earnings upside (P = 35%)

\$65B 10-year capex plan fully approved; grid hardening + EV infrastructure + smart meters; rate base grows 8% annually

Driver: Rate base growth; capex approval in IRP process

Falsifier: Capex disallowance; stranded asset risk on new generation

SCENARIO 8: EV charging infrastructure drives residential demand (P = 30%)

NC/FL EV penetration rises to 20%+ by 2028; Duke's residential load grows 1.5% above baseline; EV tariff programmes approved

Driver: EV registration data; Duke EV tariff programme results

Falsifier: EV adoption slower than forecast; managed charging reduces peak load

SCENARIO 9: Acquisition target for foreign utility (Hydro-Quebec, etc.) (P = 8%%)

Canadian utility acquires Duke for regulated US utility exposure; premium 20-30% over market; \$130-160 acquisition price

Driver: FERC/CFIUS M&A regulatory changes; utility M&A activity

Falsifier: CFIUS blocks foreign acquisition of nuclear-adjacent utility

SCENARIO 10: Bull case: load growth + nuclear + datacentre = \$150 by 2028 (P = 15%%)

All three drivers fire: datacentre load 4GW+ by 2028; nuclear relicensing approved; rate cases constructive; EPS \$8.50; stock \$148

Driver: Multi-year EPS trajectory; rate case docket; NRC filings

Falsifier: Any one driver misses; regulatory lag suppresses earnings recovery

PART VIII — SELDON VARIABLE DEEP DIVE

Variable	DUK Impact	Direction	Magnitude
K	Grid powers AI compute = knowledge capital foundation	↑ Moderate	K+ via datacentre load
ξ	8.4M customers across Southeast = energy equity anchor	↑ High	ξ++ regulated universal access
Ω	Regulated monopoly reduces energy supply risk	↓ Reduces	Ω- (reliability mandate)
I	Rate-regulated utility with bipartisan political support	↑ Mild	I+ moderate

PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
Rate case denial/ROE compression	MEDIUM (20%)	HIGH	Diversified multi-state jurisdiction
Interest rate sensitivity	MEDIUM (30%)	HIGH	Dividend growth offsets yield compression

Nuclear relicensing risk	LOW (10%)	CRITICAL	NRC track record supportive; Duke ops excellence
Capex overrun on new generation	MEDIUM (20%)	HIGH	Regulated cost-of-service model recovers overruns
Datacentre load disappointment	LOW (15%)	MEDIUM	Load growth already underway; queue visible

CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: BUY — AI DATACENTRE LOAD GROWTH UNDERPRICED IN UTILITY MULTIPLE

Duke Energy at \$124.590 (mkt cap \$97.143B) is the highest-quality way to own US AI datacentre electricity demand without the technology risk. The Research Triangle and Charlotte corridor are the fastest-growing datacentre markets in the Southeast — served exclusively by Duke. Nuclear baseload fleet provides 10 GW of firm carbon-free power at lowest marginal cost. 3.42% dividend yield + 4-5% growth = 7-9% total return. Primary risk: regulatory timing on rate cases. Seldon ξ++ (universal energy access for 8.4M customers). HUMAN_ONLY_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,555ch.